Daniel Buehrig
801504258
ITCS-3162

Week 2 Report

Initial Statistical View

The initial table comes with 10 columns. The first two are the category and description of the item, labeled as strings. Grouping by category, there are 1,181 categories across 7,413 items. Some items are described as having no category, upon further inspection, 198 items are listed in this category, or approximately 2.67% of all items. The majority of the categories are binned with counts between 1 and 30, with the higher up options sometimes going between 30 and 200, which can be seen (somewhat difficultly) in the provided bar chart. The most common categories in the product list are beef (457), pork (304), cereals RTE (291), baby food (221), and a tie between soup and no category (198).

The remaining columns are numerical columns holding the nutritional data of the product, categorized into carbohydrates, fiber, kilocalories, protein, sugar, water, saturated fat, and total lipids. Kilocalories are stored as an integer, while the rest are floats. Each is noted with a value of their specific amount, or 0 if they are not present in the item. More specific parameters will be present in the following sections.

No missing values have been found in the dataset. While some duplicate categories have been identified, their nutritional values differ, likely due to the difference in nutrition between different brands or even different products of the same general brand. Such duplicates will be monitored and documented in development of the algorithm.

Histogram and Boxplot Observations

Unedited data was split into 8 different histograms, each representing one of the 8 nutrients. 5 of the nutrients - carbohydrates, fiber, total sugar, saturated fat, and total lipids - presented a significant right-skewed histogram on their observations, inference that most of the items available have very low amounts of the product. 2 nutrients - kilocalories and protein - were also skewed to the right, but were at less extreme deviations, with kilocalories having more of a middle-right tail than protein. Water showed the most interesting plot, having a binomial dispersion, splitting items between having high or low water.

A boxplot was developed for each of the nutrients. They were not split based on their category, as the significant number of them would make even observation of one extremely narrow, so this is a combination of all categories, which will be a source of bias that will be noted. Kilocalories had been observed as being the largest boxplot, to the point that the smaller plots were being squished, so a second boxplot excluding kilocalories has also been provided. Most boxes have been observed having low medians and boxes, with somewhat high whiskers, with the three notable exceptions being the aforementioned kilocalories, carbohydrates, which has a notably larger box, and water, which has a higher median box. All have been noted with significant positive outliers, which will no doubt have to be lowered during data processing to prevent issues.

Correlation Matrix

All numerical values have been linearly correlated into a matrix and presented through a heatmap, splitting between one item and an associative item, with 7,411 degrees of freedom, as a means of evaluating appearances between two nutrients. Strong positive relationships have been observed between carbs and sugar (0.6), kilocalories and saturated fat (0.613), kilocalories and

total lipids (0.806), and saturated fat and total lipids (0.7623). Strong negative relationships exist between carbs and water (-0.757) and kilocalories and water (0.891). The remaining nutrient correlations vary between no, weak, or medium strength, with either positive or negative correlation (positive appears slightly higher).